Spout to NDI

https://spout.zeal.co

Convert between Spout and Newtek NDI sources

Version 2.030

"*Spout to NDI*" is a set of programs that allow Spout senders and receivers to share video, not only with each other but also by way of a network using the NDI ("Network Device Interface") protocol.

For more information about the NDI protocol see https://ndi.video/. This link is also available at any time by selecting Help > About.

1. Installation

Unzip "*SPOUTtoNDI_2030.zip*" in any convenient folder.

Folders

SpoutToNDI\bin – contains the required programs "*Spout to NDI.exe*" and "*NDI to Spout.exe*".

Runtime libraries

Visual Studio

Spout to NDI programs and require the runtime libraries for Visual Studio 2022 and you may be prompted to install them.

However, if you find that the programs still report missing dll's, you may need to install the Visual Studio runtime manually.

Download from here. Scroll down to find "*Other Tools, Frameworks and Redistributables*", then "*Microsoft Visual C++ Redistributable for Visual Studio 2022*". Download and install the x64 version.

NDI

The programs depend upon NDI library functions. This library "dll" file is included in the "bin" folder for the version of NDI available at the time of distribution.

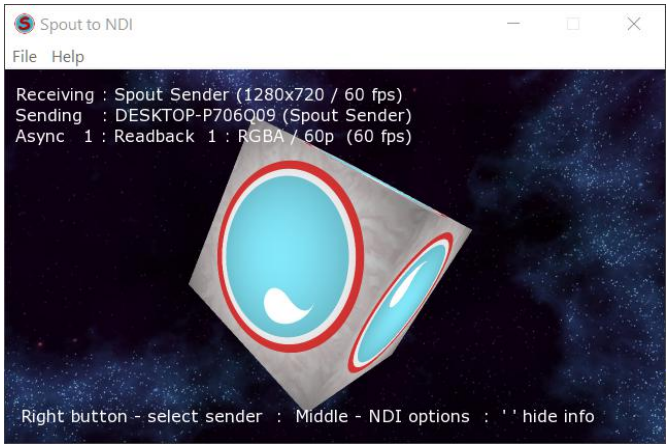
Normally you will not need to worry about it. However, if the NDI library is subsequently updated and you want to use the new version, you can install the NDI runtime.

After the runtime installation, delete the NDI dll in the distribution "bin" folder. The programs will then load the installed version. If you encounter any problems, restore the distribution dll file.

2. Operation

2.1 SPOUT to NDI

Start a Spout sender, such as the demo sender provided with the Spout installation.



Open "*Spout to NDI.exe*", located in the "bin" folder. You might receive a warning about Firewall access because it is necessary for data to be transferred across a network.

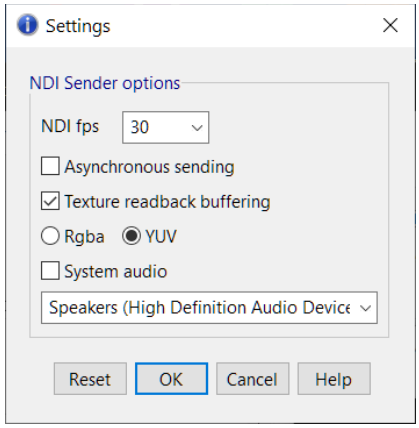
Allow access as you require. The demo Spout sender will be immediately selected. Subsequently use "File > Select sender" or press the right mouse button to select other senders.

An NDI sender is produced with the the name of the Spout sender, preceded by the name of your computer as it is identified on the network and is available to any NDI receiver on the network.

To confirm this, open the NDI Tools "Video Monitor" either on the local computer or on another computer on the same network.

Settings

Open the NDI sender options dialog by using "File > Settings" from the menu or by pressing the Middle mouse button.



Fps

The frames per second you wish to send to NDI, e.g. 29.97, 25 etc. Select from the drop-down list or enter the frame rate required and click the button to the right to submit the entry. The sender will be received at that rate.

Asynchronous sending

Activates an asynchronous sending mode where frames are sent without waiting until the previous one has been processed. The NDI system manages the buffering and synchronisation. This is the fastest sending method. Although frame timing is more accurate when disabled, the sender is also clocked at the selected frame rate and rendering is limited to that rate.

Texture readback buffering

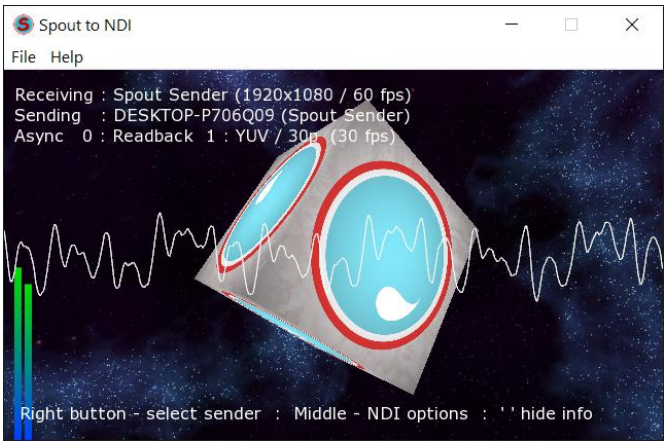
Activates asynchronous readback for copy of image data from GPU to CPU by way of OpenGL pixel buffers for increased transfer speed. Keep in mind that memory usage is increased and can become significant if multiple instances are running.

RGBA / YUV

Changes the NDI sender video frame data format. Application format is RGBA and highest quality. YUV is a compressed format with decreased chroma resolution, but it more speed efficient. The difference is noticeable at high resolutions such as 4K (3840x2160) and can achieve 40-50 fps at instead of 20-30 fps for RGBA.

System audio

Activates capture of system audio and sends it together with image data. The system audio device can also be selected. The space bar disables or enables the audio waveform display, but leaves the bar graph to show that audio is being captured.



Starting from a command line

"*Spout to NDI.exe*" can be started from a command line in order to specify the Spout sender to receive from. Change to the "bin" folder containing the executable. Open a command console (cmd.exe). A batch file "aa-dos.bat" in the bin folder will do that. Specify the sender to receive from. For example :

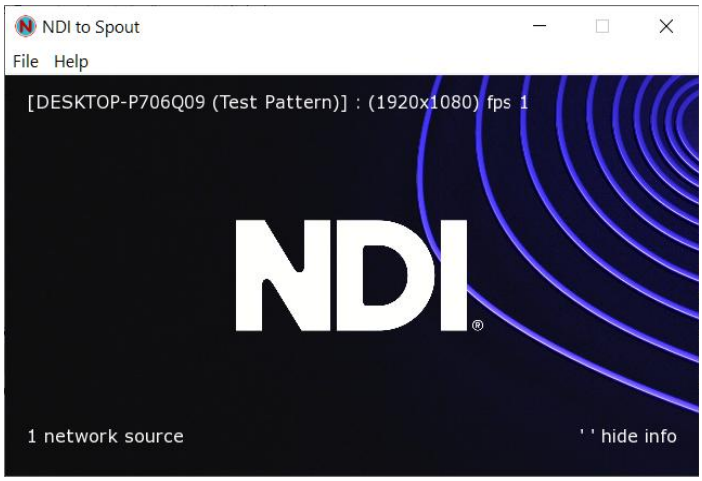
"Spout to NDI.exe" "spout sender to receive from"

Double quotes are removed from the name. The application will start minimized and wait for the selected sender to open. When it opens, the NDI output will start. There is a slight lag due to the NDI processing. If the sender closes, the program will wait for it to open again. The option is disabled if the user selects another sender.

An example batch file "aa-spout-sender.bat" will receive from "Spout Demo Sender". You can modify this as required.

2.2 NDI to SPOUT

NDI to Spout receives from any NDI sender and produces a Spout sender which can then be used to share textures at high speed between applications on the receiving machine.

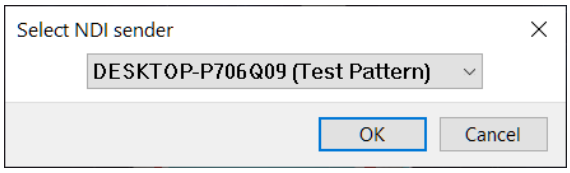


Open "Test Patterns" from the NDI tools so that an NDI sender is available.

Then open "*NDI to Spout.exe*", located in the "bin" folder. This can be started on any machine on the network but for testing open it on the source machine. Now the NDI sender test pattern is being received and the output is a Spout sender of the same name preceded by the name of your computer as identified on the network. Run the Spout demo receiver and you will see it listed. It will also be available in any program that supports Spout input.

Selecting a sender

"File > Select sender" or right mouse click to bring up a dialog with a list of NDI senders currently available on the network. Drop down the list to select the one you want.



Sometimes the network may be temporarily unavailable and senders might disappear and re-appear in the list. This will not affect the selected NDI sender and when that sender comes back on line, the frames will resume. It may take some seconds for discovery of all the NDI senders on the network.

Low bandwidth

"File > Low bandwidth" specifies whether the receiver connection is in high or low bandwidth mode. "Low" provides a medium quality progressive stream that takes almost no bandwidth. This is normally about 640 pixels in size on the longest side. "High" results in the same stream that is being sent from the up-stream source.

Audio

“File > Audio” allows audio to be received from the NDI source as well as the graphics and plays through the speakers. Audio is not sent to the Spout receiver. The space bar disables or enables the audio waveform display, but leaves the bar graph to show that audio is being received.

Starting from a command line

“*NDI to Spout.exe*” can be started from a command line in order to specify the NDI sender to receive from. Change to the “bin” folder containing the executable. Open a command console (cmd.exe), or use “aa-dos.bat”. Specify the sender name to receive from.

For example :

"NDI to Spout.exe" "NDI sender to receive from"

Double quotes are removed from the name.
NDI sender names are combined from two parts :

The computer e.g. DESKTOP-P706Q09
The sender in brackets e.g. (Test Pattern)
To make the format : DESKTOP-P706Q09 (Test Pattern)

"NDI to Spout.exe" "DESKTOP-P706Q09 (Test Pattern)"

The application window will minimize and wait for the selected sender to open. There is a slight delay due to the NDI processing. When it opens, the Spout output will start. If the selected sender closes, the application will wait for it to open again. This is normal behaviour as explained above. The command line option allows the application to start minimised and wait for the required sender to open instead of connecting to the first in the list. The example batch file “aa-ndi-sender.bat” will receive from the NDI Tools application “Test Patterns”, which sends with the application name "Test Pattern". The computer name is unique for the particular computer, in this case "DESKTOP-P706Q09". The complete name is :

"DESKTOP-P706Q09 (Test Pattern)"

The batch file must be modified for the name of the computer you wish to receive from. Note that "Studio Monitor" precedes the name with a prefix "NDI - ". Do not include this prefix.

2.3 Magic

"Magic*NDIsender.dll*" and "Magic*NDIreceiver.dll*" are plugins for Magic Music Visuals for sending to and receiving from NDI applications.

They can be found on the GitHub repository and downloaded from the releases page.

Magic does not have NDI support at the time of writing but may in the future, so these plugins could be useful in the meantime.

The modules are both 64bit and tested for compatibility with Magic Version 2.33. Refer to the documentation in the repository.

2.4 NDISyphon

"NDI*Syphon*" is the equivalent of "*Spout to NDI*" but for OSX using *Syphon* and has been developed as a free utility by VidVox.

OSX machines using "NDI*Syphon"* can receive textures from any application with *Syphon* support and share video over a network. Requirements are OSX 10.8+.

On a Mac, start any application with *Syphon* output. Then start "*NDISyphon"* and select the *Syphon* source. The *Syphon* texture will then be transmitted over the network using the Newtek NDI protocol and any machine on that network can access the video.

An OSX receiving machine can then use "*NDISyphon"* to select an NDI source and broadcast this as a *Syphon* texture to any applications on that machine with *Syphon* input.

A Windows receiving machine can use *"NDI to Spout"* to select an NDI source and broadcast this as a *Spout* texture to any applications on that machine with *Spout* input.

Download from here.

2.5 NDI tools

Here is a download link for useful tools. After entering your details, you will receive an email with a link to download them.

These include a "Test Pattern" sender, a "Video Monitor" to receive from NDI senders and a "Scan Converter" screen capture application.

3. Licencing

3.1 Spout to NDI

"Spout to NDI" is released under the Simplified BSD licence.

Copyright (C) 2016-2026. Lynn Jarvis, https://spout.zeal.co/, All rights reserved.

Redistribution and use in source and binary forms, with or without modification, are permitted provided that the following conditions are met:

1. Redistributions of source code must retain the above copyright notice, this list of conditions and the following disclaimer.

2. Redistributions in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer in the documentation and/or other materials provided with the distribution.

THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED.

IN NO EVENT SHALL THE COPYRIGHT HOLDER OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

3.2 NDI

NDI ("Network Device Interface") is copyright Vizrt NDI AB. (https://ndi.video/).